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Prepared in cooperation with the Massachusetts Department of Environmental Protection, the U.S. Environmental Protection Agency, and the City of Cambridge

Potential Reductions of Street Solids and Phosphorus in Urban Watersheds from Street Cleaning, Cambridge, Massachusetts, 2009–11

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Abstract

Material accumulating and washing off urban street surfaces and ultimately into stormwater drainage systems represents a substantial nonpoint source of solids, phosphorus, and other constituent loading to waterways in urban areas. Cost and lack of usable space limit the type and number of structural stormwater source controls available to municipalities and

other public managers. Non-structural source controls such as street cleaning are commonly used by cities and towns for construction, maintenance and aesthetics, and may reduce contaminant loading to waterways. Effectiveness of street cleaning is highly variable and potential improvements to water quality are not fully understood. In 2009, the U.S. Geological Survey, in cooperation with the Massachusetts Department of Environmental Protection, the U.S. Environmental Protection Agency, and the city

First posted February 12, 2013

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This is an electronic copy of Appendix 1 that contains tables 1–1 through 1–6

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of Cambridge, Massachusetts, and initiated a study to better understand the physical and chemical nature of the organic and inorganic solid material on street surfaces, evaluate the performance of a street cleaner at removing street solids, and make use of the Source Loading and Management Model (SLAMM) to estimate potential reductions in solid and phosphorus loading to the lower Charles River from various street-cleaning technologies and frequencies.

Average yield of material on streets collected between May and December 2010, was determined to be about 740 pounds per curb-mile on streets in multifamily land use and about 522 pounds per curb-mile on commercial land-use streets. At the end-of-winter in March 2011, about 2,609 and 4,788 pounds per curb-mile on average were collected from streets in multifamily and commercial land-use types, respectively. About 86 percent of the total street-solid yield from multifamily and commercial land-use streets was greater than or equal to 0.125 millimeters in diameter (or very fine sand). Observations of street-solid distribution across the entire street width indicated that as much as 96 percent of total solids resided within 9 feet of the curb. Median accumulation rates of street solids and median washoff of street solids after rainstorms on multifamily and commercial land-use streets were also similar at about 33 and 22 pounds per curb-mile per day, and 35 and 40 percent, respectively. Results indicate that solids on the streets tested in Cambridge, Mass., can recover to pre-rainstorm yields within 1 to 3 days after washoff. The finer grain-size fractions tended to be more readily washed from the roadway surfaces during rainstorms. Street solids in the coarsest grain-size fraction on multifamily streets indicated an average net increase following rainstorms and are likely attributed to debris run-on from trees, lawns, and other plantings commonly found in residential areas.

In seven experiments between May and December 2010, the median removal efficiency of solids from street surfaces following a single pass by a regenerative-air street cleaner was about 82 percent on study sites in the multifamily land-use streets and about 78 percent on the commercial land-use streets. Median street-solid removal efficiency increased with increasing grain size. This type of regenerative-air street cleaner left a median residual street-solid load on the street surface of about 100 pounds per curb-mile.

Median concentrations of organic carbon and total phosphorus (P) on multifamily streets were about 35 and 29 percent greater, respectively, than those found on commercial streets. The median total mass of organic carbon and total P in street solids on multifamily streets was 68 and 75 percent greater, respectively, than those found on commercial streets. More than 87 percent of

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the mass of total P was determined to be in solids greater than or equal to 0.125 millimeters in diameter for both land-use types. The median total accumulation rate for total P on multifamily streets was about 5 times greater than on commercial streets. Total P accumulation in the medium grain-size fraction was nearly the same for streets within both land-use types at 0.004 pounds per curb-mile per day. Accumulation rates within the coarsest and finest grain-size fractions on multifamily streets were about 11 and 82 times greater than those on the commercial streets. Median washoff of total P was 58 and 48 percent from streets in multifamily and commercial land-use types, respectively, and generally increased with decreasing grain size. Total P median reductions resulting from a single pass of a regenerative-air street cleaner on streets in multifamily and commercial land-use types were about 82 and 62 percent, respectively, and were similar in terms of grain size between both land-use types.

A Source Loading and Management Model for Microsoft Windows (WinSLAMM) was applied to a 21.8 acre subcatchment in Cambridge, Mass. The subcatchment area consists of mostly commercial and multifamily land-use types to evaluate the potential reductions of total and particulate solids, and P attributed to street cleaning. Rainwater runoff from rooftops represented between 20 and 50 percent of the total basin runoff. Street surfaces only accounted for about 20 percent of the total basin runoff.

Monthly applications of mechanical-brush and vacuum-assisted street cleaners within the subcatchment as defined by SLAMM for areas with long-term (24-hour) on-street parking and monthly parking controls using five average climatic years resulted in total solid reductions of about 3 and 5 percent, respectively. Simulating the regenerative-air street cleaner tested as part of this study resulted in total solid reductions of about 16 percent. Increasing street cleaning frequency to three times weekly increased total solids removal for mechanical-brush, vacuum-assisted, and regenerative-air street cleaners to about 6, 14, and 19 percent, respectively. Monthly applications of mechanical-brush, vacuum-assisted, and regenerative-air street cleaners within the subcatchment resulted in total P reductions of about 1, 3, and 8 percent, respectively. A street cleaning frequency of three times each week for each of the three street-cleaner types increased total P removal to about 3, 7, and 9 percent, respectively.

Suggested citation:

Sorenson, J.R., 2013, Potential reductions of street solids and phosphorus in urban watersheds from street cleaning, Cambridge, Massachusetts, 2009–11: U.S. Geological Survey Scientific

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Page Last Modified: Thursday, 07-Mar-2013 09:41:23 EST

